

NM_024596.5(MCPH1):c.2254C>G (p.Arg752...

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Germline

Classification
★ ★ ★ ★
Uncertain significance
Uncertain significance (2)
2 out of 2 submissions contributed to this classification

Somatic

No data submitted for somatic clinical impact

Somatic

No data submitted for oncogenicity

Variant Details

Variant details

Identifiers:

NM_024596.5(MCPH1):c.2254C>G (p.Arg752Gly)

Type and length:

single nucleotide variant, 1 bp

Location:

Variation ID: 908237

Accession: VCV000908237.5

Cytogenetic: 8p23.1

8: 6621493 (GRCh38) [NCBI UCSC]

8: 6479014 (GRCh37) [NCBI UCSC]

Timeline in ClinVar:

HGVS:

First in ClinVar	Last submission	Last evaluated
Germline	May 31, 2020	Feb 7, 2023
		Aug 9, 2022

Nucleotide	Protein	Molecular consequence
NM_024596.5:c.2254C>G	MANE SELECT	NP_078872.3:p.Arg752Gly
NM_001322042.2:c.2254C>G		NP_001308971.2:p.Arg752Gly
NM_001363979.1:c.2254C>G		NP_001350908.1:p.Arg752Gly

Protein change: R752G, R659G

Other names: -

... more HGVS

Canonical SPDI: NC_000008.11:6621492:C:G

Global minor allele frequency (GMAF) : 0.00020 (T)

Allele frequency : Trans-Omics for Precision Medicine (TOPMed) 0.00003

Links :

ClinGen: CA4611478

dbSNP: rs373658774

VarSome

Genes

Gene	OMIM	ClinGen Gene Dosage Sensitivity Curation		Variation Viewer	Related variants	
		HI score	TS score		Within gene	All
MCPH1		Gene associated with autosomal recessive phenotype	Not yet evaluated	GRCh38 GRCh38 GRCh38 GRCh37	986	1545
MCPH1-AS1	-	-	-	GRCh38 GRCh38	-	255

Conditions - Germline

Condition	Classification (# of submissions)	Review status	Last evaluated	Variation/condition record
Microcephaly 1, primary, autosomal recessive	Uncertain significance (1)	★ ★ ★ ★	Jan 12, 2018	RCV001158612.4
not provided	Uncertain significance (1)	★ ★ ★ ★	Aug 9, 2022	RCV002558400.2

Submissions - Germline

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows	
Uncertain significance (Jan 12, 2018)  Contributing to aggregate classification	★★★★ (ICSL Variant Classification Criteria 13 December 2019)	Microcephaly 1, primary, autosomal recessive	Illumina Laboratory Services, Illumina Accession: SCV001320261.1 First in ClinVar: May 31, 2020 Last updated: May 31, 2020		

Comment:
This variant was observed in the ICSL laboratory as part of a predisposition screen in an ostensibly healthy population. It had not been previously curated by ICSL or reported in the Human Gene Mutation Database (HGMD: prior to June 1st, 2018), and was therefore a candidate for classification through an automated scoring system. Utilizing variant allele frequency, disease prevalence and penetrance estimates, and inheritance mode, an automated score was calculated to assess if this variant is too frequent to cause the disease. Based on the score, this variant could not be ruled out of causing disease and therefore its association with disease required further investigation. A literature search was performed for the gene, cDNA change, and amino acid change (if applicable). No publications were found based on this search. This variant was therefore classified as a variant of unknown significance for this disease. ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Uncertain significance (Aug 09, 2022)  Contributing to aggregate classification	★★★★ (Invitae Variant Classification Sherlock (09022015))	not provided	Labcorp Genetics (formerly Invitae), Labcorp Accession: SCV003275994.1 First in ClinVar: Feb 07, 2023 Last updated: Feb 07, 2023		
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Comment:
This sequence change replaces arginine, which is basic and polar, with glycine, which is neutral and non-polar, at codon 752 of the MCPH1 protein (p.Arg752Gly). This variant is present in population databases (rs373658774, gnomAD 0.01%). This variant has not been reported in the literature in individuals affected with MCPH1-related conditions. ClinVar contains an entry for this variant (Variation ID: 908237). Algorithms developed to predict the effect of missense changes on protein structure and function are either unavailable or do not agree on the potential impact of this missense change (SIFT: "Not Available"; PolyPhen-2: "Benign"; Align-GVGD: "Not Available"). In summary, the available evidence is currently insufficient to determine the role of this variant in disease. Therefore, it has been classified as a Variant of Uncertain Significance. ([less](#))

Observation 1

Collection method: clinical testing
Allele origin: germline
Affected status: unknown

Citations for germline classification of this variant

There are no citations for germline classification of this variant in ClinVar. If you know of citations for this variation, please consider submitting that information to ClinVar.

Text-mined citations for rs373658774

These citations are identified by LitVar using the rs number, so they may include citations for more than one variant at this location. Please review the LitVar results carefully for your variant of interest.

Record last updated Apr 13, 2026 

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